

resonant converter design, because of its high-power and high-efficiency characteristics, to obtain DC power for charging the battery.

The design utilises a rectifier circuit for converting input AC voltage to high-voltage DC output, and it also has an electromagnetic interference (EMI) filter to eliminate high-frequency noise from input power source. A pulse-width modulation (PWM) controller IC, like UC3525, can be used for driving the MOSFETs of half-bridge LLC converter.

The battery charging process is supervised by the MCU HT45F5Q-2. It monitors the battery voltage and charging current levels and gives feedback to the PWM controller IC. Based on the feedback, the PWM controller varies the duty cycle of its PWM signal and drives the MOSFET circuit to obtain variable output voltage and current for charging the battery.

For better protection, HT45F5Q-2 is isolated from rest of the circuit (i.e., high-voltage components) using a photo-coupler. Battery-level LED indicators are provided for knowing the charging status.

Battery charging process

The change in charging voltage and current during the charging process is graphically illustrated in Fig. 4. If the battery voltage is too low when connected for charging, low charging current (i.e., trickle charge (TC)) will be set initially and charging process will start.

When the battery voltage increases to a pre-defined level (V_u), constant voltage (CV) and constant current (CC) is applied for charging and continued until the battery is fully charged. Battery is considered to be fully charged when voltage reaches V_{OFF} . When charging current drops to I_u , final voltage (FV) is set. The voltage, current and temperature control process in this EV charger are explained below.

(a) Voltage control. The charging voltage is decided based on the initial



Fig. 6: EV charger PCB assembly

voltage of battery when it is connected for charging. As the charging progresses, charging voltage changes accordingly and, finally, when battery is fully charged, the final voltage is set. The charging-voltage decision levels for 48V/12A battery charger are explained below.

- If Battery Voltage < 36V, TC(0.6A) Charging, Voltage Setting FV(56V)
- If Battery Voltage < 40V, TC(0.6A) Charging, Voltage Setting CV(58V)
- If Battery Voltage > 40V, CC(12.0A) Charging, Voltage Setting CV(58V)
- When fully charged, voltage is set to FV(56V). If battery voltage is lower than FV, the charging current will be reset to CC (12.0A).

(b) Current control. Charging current is set depending on the battery voltage. Initially, if the battery voltage is too less, trickle-charge current would be set for charging the battery. Once battery voltage reaches certain level, constant current is supplied for charging, until battery is charged fully. The charging-current decision levels for 48V/12A battery charger are listed below.

- Recharging Current < 1.2A, determine the end of charging
- Recharging Current > 0.2A, determine the start of charging

(c) Over-temperature protection. The EV charger has a negative temperature coefficient (NTC) thermistor to monitor the temperature and a fan to regulate the heat. When temperature increases, the fan is automatically switched on to dissipate the

heat; it gets switched off when the temperature is reduced to the lower set threshold. Also, the fan turns on when charging current is high and turns off when charging current is low.

- When NTC temperature > 110°C, the charging current will be reduced to 50 per cent of charging current and will be monitored periodically

(d) LED indications for charging status. These are listed below.

- TC charge, red light flashes slowly (0.3 sec on, 0.3 sec off)
- CC, CV charge, red light flashes quickly (0.1 sec on, 0.1 sec off)
- When not charging, green light is on
- When charging time exceeds eight hours, red and green lights are bright

(e) Charging duration. When charging duration is exceeded (duration depends on battery capacity), the voltage drops to FV, the current is reduced to TC, and charger repeatedly monitors the battery voltage.

Schematic and PCB assembly

The schematic of Holtek EV charger design for 48V/12A type is shown in Fig. 5 for reference and its PCB assembly is shown in Fig. 6.

The ASSP flash MCU HT45F5Q-2 can also be used for designing higher-wattage solutions. It offers a programmable option for setting parameter thresholds, which makes it very convenient for EV charger designs. Holtek provides technical resources such as block diagram, application circuits, PCB files, source code, etc to help designers in rapid product development and speed up time-to-market.

EV charger development platform for HT45F5Q-X series will also be available soon. Using this software tool, users would be able to easily select the charging voltage/current and other parameters to create a program. This application will also be able to generate a program containing a standard charging process, thereby significantly simplifying the development process. **EFY**